
BioModel Optimizer

Anthony Di Pietro^{* 1} Sara Lazzaroni^{* 1}

Abstract

Dynamic simulation of biological pathways requires accurate parameterization of reaction rates and kinetic constants, which are often missing from standard pathway databases. We present a novel, automated workflow that combines Large Language Models (LLMs) with high-performance numerical simulation to estimate and fit missing biological parameters in Systems Biology Markup Language (SBML) models. Our approach leverages the latent biological knowledge embedded in state-of-the-art LLMs to establish baseline target concentrations for pathway species. We then frame parameter estimation as a continuous optimization problem, using a derivative-free Evolutionary Strategy to align the pathway’s simulated behavior with these targets. This hybrid approach demonstrates a scalable framework for bridging the gap between qualitative topological models and quantitative dynamic simulations.

1. Introduction

Mathematical modeling of biochemical networks is a cornerstone of systems biology, enabling researchers to simulate complex cellular behaviors. However, while databases like Reactome (Ragueneau et al., 2026) provide high-quality topological networks encoded in SBML, they frequently lack the specific kinetic parameters required for dynamic simulations. Manually curating these parameters from literature is time-consuming and often yields disparate values obtained under incompatible experimental conditions.

To address this bottleneck, we developed an automated parameter-fitting pipeline that utilizes Ollama, a framework for executing locally-hosted Large Language Models (LLMs), to generate realistic target values for molecular species given SBML metadata. Subsequently, `libRoadRunner` is employed alongside an OpenAI Evolution Strategies (OpenAI-ES (Salimans et al., 2017)) optimization algorithm to iteratively tune uncharacterized model parameters (e.g., binding constants) until the simulated terminal concentrations match the LLM-derived targets. To ensure biological robustness, we propose a two-stage evaluation framework: optimization is first performed over a shorter simulation horizon, followed by evaluation against a larger test-set horizon to verify that the species concentrations have achieved a true, stable steady-state. The optimized parameters are then programmatically written back into the SBML architecture.

2. Background

2.1. Systems Biology Markup Language (SBML) and Reactome

The Systems Biology Markup Language (SBML) (Hucka et al., 2003) is a XML-based standard for representing computational models of biological processes. While a biological pathway might intuitively be visualized as a simple flowchart of interacting molecules, SBML strictly formalizes this topology into a machine-readable architecture. It defines the biological environment through a hierarchy of discrete components: *compartments* (the physical spaces where reactions occur, such as the cytosol or nucleus), *species* (the biochemical entities, like proteins or metabolites, residing in those compartments), and *reactions* (the stoichiometric rules and kinetic laws governing how species are consumed, produced, or modified).

The models utilized in this study are sourced from Reactome (Ragueneau et al., 2026), a highly curated, peer-reviewed database of biological pathways and processes. When Reactome pathways are exported as SBML files, they provide

^{*}Equal contribution ¹Department of Computer Science, Sapienza University of Rome, Rome, Italy. Correspondence to: Anthony Di Pietro <dipietro.1960447@studenti.uniroma1.it>, Sara Lazzaroni <lazzaroni.1983548@studenti.uniroma1.it>.

high-fidelity topological networks. However, they do not natively include the mathematical kinetic laws, namely the explicit equations that govern how species interact over time, nor the quantitative parameters, such as reaction rates and binding affinities, that are required to perform dynamic time-course simulations.

2.2. Ollama and Local LLMs

Ollama is a lightweight, open-source framework designed to easily run, manage, and deploy Large Language Models (LLMs) entirely locally. By avoiding reliance on remote cloud APIs like Gemini or OpenAI, Ollama guarantees complete data privacy and eliminates external network dependencies, rate limits, and recurring API costs. Within our automated pipeline, Ollama provides the foundational infrastructure to reliably query advanced conversational models directly on algorithmic modeling hardware, deducing complex biological ground truths from raw SBML metadata efficiently and reproducibly.

2.3. libRoadRunner

libRoadRunner is a highly efficient simulation engine for SBML models (Welsh et al., 2022). It is able to perform time-course simulations and steady-state analyses, making it ideal for the iterative evaluations required in machine learning loops. During a simulation, the engine automatically compiles the discrete SBML reactions into a continuous system of ordinary differential equations (ODEs). It inherently aggregates all the production and consumption terms for every species based on the defined reaction stoichiometries and kinetic laws. Consequently, the total rate of change for any species is automatically computed without requiring explicit global mass-balance equations. This means we simply define the individual reactions-such as continuous inputs (`reaction_input`) or output degradation (`reaction_output_degradation`)-and the engine automatically tracks the overall accumulation and depletion of each species over time.

2.4. OpenAI Evolution Strategies (OpenAI-ES)

OpenAI-ES (Salimans et al., 2017) is a scalable, derivative-free optimization algorithm. Unlike traditional gradient descent, which requires differentiable objective functions, ES estimates the gradient of a smoothed objective by sampling parameters perturbed by Gaussian noise. This makes it highly robust to the rough, non-convex loss landscapes typical of biological dynamic systems.

3. Dataset & Methods

3.1. Pathway Merging

Before generating target concentrations or augmenting the files with kinetic rules, our pipeline supports merging multiple distinct SBML pathways into a unified model. This allows for the dynamic simulation and optimization of more complex biological processes. The merging procedure programmatically fuses compartments, chemical species and reactions from two source SBML files by systematically resolving conflicts and deduplicating identical entities.

To guarantee a consistent topological framework, the integration logic relies on explicit equivalence criteria for each biological component:

- **Compartments** are merged strictly based on exact matching of their string identifiers (the `id` attribute).
- **Species** are evaluated and deduplicated using a composite key consisting of their `id` and their assigned compartment.
- **Reactions** undergo a rigorous topological and structural comparison. Beyond their basic IDs, reaction equivalence is established through a unique signature that encompasses the sequence of participating compartments, strictly sorted lists of reactants, products and modifiers alongside their precise stoichiometries, the reversibility flag and the mathematical formula of any pre-existing kinetic law.

This merged model subsequently serves as the consolidated input for all downstream kinetic augmentation and parameter optimization steps.

3.2. Dataset Preparation

The methodology was developed and tested using human biological pathway models sourced from the Reactome (Ragueneau et al., 2026) database. When exported to the standard SBML format, Reactome provides a rigorous topological framework

detailing the hierarchy of cellular compartments, the inventory of chemical species and the exact stoichiometries of their interactions. However, because Reactome natively lacks quantitative dynamics, a crucial preprocessing step in our dataset preparation involves augmenting these topological models. Furthermore, raw SBML exports frequently contain non-canonical or prefixed MathML namespace declarations (e.g., `<ns7:math>`) that trigger structural validation errors in strict parsers like `libSBML`. To guarantee robustness, our pipeline first applies a sanitization step that canonicalizes all MathML namespaces across the XML document. Subsequently, we systematically insert mathematical kinetic rate laws for the pathway’s reactions. While this establishes the structural ordinary differential equations (ODEs) required for simulation, the exact numerical parameter values (such as binding affinities and specific reaction rates) remain unknown and must be fitted.

To accurately model continuous biological processes, the species dynamics were mathematically augmented with specific generation and degradation rules. For any given input species x_i , its dynamic behavior was defined as:

$$\frac{dx_i}{dt} = K_i - \sum \text{consumers}(x_i)$$

Conversely, for output species, the model behavior was adjusted to reflect continuous removal:

$$\frac{dx_i}{dt} = \sum \text{producers}(x_i) - K_i \cdot x_i$$

where K_i represents the unknown rate constants.

Additionally, to properly evaluate the system’s behavior over time and avoid instantaneous fluctuations, we augmented the SBML models with rate rules to compute the running time-averages of the species concentrations and their variance. For a species concentration x , its running mean y is governed by the differential equation:

$$\frac{dy}{dt} = \frac{x - y}{t + \epsilon}$$

where ϵ (e.g., 10^{-6}) is a small constant to prevent division by zero at $t = 0$. This ensures that $y(t) = \frac{1}{t} \int_0^t x(\tau) d\tau$. Similarly, the second-order moment y_2 is tracked via:

$$\frac{dy_2}{dt} = \frac{x^2 - y_2}{t + \epsilon}$$

These running moments allow us to directly compute the variance $\text{Var}(x) = y_2 - y^2$ and the standard deviation $\text{StdDev}(x) = \sqrt{\text{Var}(x)}$ dynamically during the simulation; while the optimization objective itself is defined on the terminal state (Section 3.5), these statistics are retained as diagnostics for assessing whether a candidate trajectory has reached a stable steady state. Finally, for documentation and downstream validation, explicit constraints encoding the target reference value μ_i supplied by the LLM are appended to the SBML definitions for each species; these are recorded as model metadata and are not enforced by the optimizer, which is driven solely by the objective of Section 3.5.

3.3. Regulatory Kinetics and Unit Consistency

Beyond the boundary generation and degradation terms, many Reactome reactions are governed by *modifier* species that regulate a reaction without being consumed. To capture this behaviour, reactions carrying modifiers are augmented with cooperative Hill terms. An activating modifier multiplies the reaction rate by

$$H^+(x, M_c) = \frac{(x/M_c)^n}{1 + (x/M_c)^n},$$

whereas an inhibiting modifier contributes

$$H^-(x, M_c) = \frac{1}{1 + (x/M_c)^n},$$

with a fixed Hill coefficient $n = 10$ that produces a sharp response. The threshold M_c is species-specific and is set to the modifier’s own reference concentration, so that the regulator acts around its physiological level. The polarity of each modifier (activating versus inhibiting) is resolved from its SBO term. Crucially, these regulatory terms are applied only to genuine modifier interactions and not to the boundary source and sink reactions, which are kept linear; this preserves a

monotonic dependence of the steady state on the boundary rate constants and avoids the highly non-convex landscapes that steep Hill terms would otherwise introduce into the optimization.

A further consideration concerns physical units. In our augmented models the kinetic laws operate on species *concentrations*, which is also the quantity reported by the LLM. In principle, one could convert each reference concentration to an *abundance* through the compartment volume, $\text{abundance}_i = \text{concentration}_i \times V_{c(i)}$, where $c(i)$ denotes the compartment of species i . In practice, however, realistic cellular volumes are on the order of $\sim 10^{-14}$ L; multiplying them through drives the resulting abundances to $\sim 10^{-30}$, which renders the ODE system numerically intractable and causes the integrator to fail. We therefore fit directly in concentration: compartment sizes are kept at unity and the optimization targets are the LLM concentrations themselves. The per-compartment volumes V_c are nonetheless retained as metadata, so an abundance can be recovered for any species *post hoc* via $\text{abundance}_i = \text{concentration}_i \times V_{c(i)}$ without affecting the fit.

3.4. LLM-Driven Target Generation

To perform parameter optimization on these mathematically augmented models, empirical target data for the system’s steady state is required. Due to the scarcity of experimental data, we developed a methodology to infer these targets using a Large Language Model.

The process begins by parsing the internal architecture of the prepared SBML model. We extract the structural topology, the newly inserted kinetic laws, and the embedded semantic metadata (specifically the RDF annotations and SBO terms). This highly contextual biological and mathematical data is synthesized into a structured prompt and processed by the LLM. By interpreting the structural equations alongside the RDF links and SBO classifications, the model leverages its latent training over biological literature to deduce the functional roles of each molecule.

To prevent context truncation or malformed JSON outputs typical of large requests, the LLM querying process is implemented using a chunked generation strategy. The species list is partitioned into smaller subsets (e.g., 30 species per chunk), and target concentrations are requested iteratively. A tolerant accumulation mechanism is employed: any species omitted by the LLM in a given chunk are retried in subsequent passes. If any species remain missing after retries, their targets are imputed using the median concentration of the successfully retrieved species. This ensures that the pipeline never aborts and a complete target dataset is robustly generated for all pathway species.

Through this procedure, the system sequentially extracts, in a structured machine-readable format, a biologically plausible mean steady-state concentration μ_i for every species together with a physical volume V_c for every compartment. As described in Section 3.3, the model is fitted directly in concentration: the LLM concentrations μ_i serve as the reference ground truth against which our optimization pipeline fits the missing parameters within the kinetic laws, while the compartment volumes V_c are retained for optional *post hoc* abundance recovery.

3.5. Dynamic Simulation and Objective Function

To evaluate candidate parameter sets, the methodology employs a high-performance numerical simulation engine capable of solving the ordinary differential equations (ODEs) defined by the SBML architecture.

For a given set of candidate parameters θ , a dynamic time-course simulation is executed from $t = 0$ to a designated terminal time T . The terminal concentration of each simulated species, denoted $s_i(\theta, T)$, is extracted. Because the reference targets μ_i span several orders of magnitude across species, a plain mean-squared error on raw concentrations is dominated by the largest targets and is numerically ill-behaved for the smallest ones. We therefore define the objective in logarithmic space:

$$F(\theta) = \sum_i (\log_{10} s_i(\theta, T) - \log_{10} \mu_i)^2.$$

This logarithmic formulation weights every species equally regardless of its magnitude and, unlike a clipped relative error, never saturates: a species that is several orders of magnitude away from its target still contributes an informative gradient rather than a flat plateau, which we found essential for reliable convergence. Because dynamic simulations of complex biological networks are computationally expensive and often produce unstable parameter sets, we further implemented a two-stage optimization and validation strategy to ensure robust steady-state convergence.

3.6. Stage 1: Derivative-Free Optimization

In the first stage, optimization is performed over a shorter, computationally efficient simulation horizon ($T_{opt} = 100,000$). We frame the parameter estimation as a continuous, derivative-free optimization problem utilizing an Evolution Strategies (ES) algorithm. The search space is parameterized in the log-domain (θ) to strictly enforce biological positivity when the parameters are exponentiated (10^θ) for the simulation.

The ES algorithm iteratively evolves a population of candidate parameter sets using mirrored sampling to reduce variance. At each step:

1. The parameter space is perturbed symmetrically using Gaussian noise, creating paired positive and negative mutations.
2. The objective function $F(\theta)$ is evaluated for each candidate at the short horizon T_{opt} . Because the evaluation of each candidate is completely independent, the algorithm leverages multiprocessing to farm the candidate simulations across a pool of isolated worker processes, significantly accelerating the main computational bottleneck.
3. These evaluations are converted into normalized maximization scores, which are used to estimate the gradient of the smoothed objective landscape.
4. A gradient-ascent step is taken to update the global parameters toward the most successful mutations.

The update is driven by an Adam optimizer with decaying-but-floored learning rate and perturbation schedules, which preserves exploration early on while permitting fine refinement near convergence. To avoid spending the full iteration budget once a satisfactory fit has been reached, the search terminates early as soon as the best objective value falls below a configurable tolerance.

3.7. Stage 2: Steady-State Validation

In dynamic biological modeling, iterative algorithms can sometimes falsely converge on parameters that temporarily intersect target values at T_{opt} but exhibit wild oscillations, divergence, or crashes at later time points.

To control for this transient behavior, Stage 2 acts as an extended validation phase. The best-performing parameter set discovered in Stage 1 is subjected to a vastly extended simulation horizon ($T_{val} = 1,000,000$, an order of magnitude beyond T_{opt}). We then assess biological robustness by inspecting whether the simulated concentrations at T_{val} remain stable, closely matching the targets and showing no substantial deviation from the state achieved at T_{opt} , rather than collapsing into oscillatory behavior or diverging over time.

Once validated as a true mathematical steady state, the optimized kinetic parameters are programmatically injected back into the underlying XML structure of the SBML model. This finalized model is thereby transformed from a static topological diagram into a fully parameter-fitted dynamic simulation, ready for downstream analysis.

4. Results

We evaluate the pipeline along two complementary axes: *scalability* (wall-clock time and peak memory of each stage as model size grows) and *correctness* (whether each stage produces valid, biologically sensible output, and whether the optimizer recovers parameters that satisfy the targets). All experiments were produced by a single automated benchmark harness that runs the full merge $\rightarrow$ augment $\rightarrow$ target $\rightarrow$ inject $\rightarrow$ optimize $\rightarrow$ simulate pipeline and records per-stage metrics.

4.1. Experimental Setup

The hardware utilized for benchmarking is equipped with an AMD Ryzen 7 9800X3D processor, an NVIDIA GeForce RTX 5080 GPU, and 64 GB DDR5 RAM. The pipeline leveraged 16 multiprocessing workers for parallel execution. Target generation was performed using the llama3.1:8b Large Language Model hosted locally via Ollama. The OpenAI-ES optimizer was configured with a population size of 20 and a maximum budget of 5000 iterations.

We consider scenarios of increasing scale, each obtained by merging curated Reactome pathways. To ensure the merge step is non-trivial, the selected pathways share at least one chemical species. Table 1 details the characteristics of the three generated benchmark models (*small*, *medium*, and *large*). For each scenario, the table lists the source Reactome stable identifiers (R-HSA) and quantifies the complexity of the resulting merged network: the total number of unique chemical species (Spec.), the number of dynamic reactions (Rxns), the final count of tunable kinetic parameters that the optimization

algorithm must fit (Par.) and the combined file size of the raw SBML inputs in megabytes (MB).

Table 1. Benchmark scenarios and aggregated model sizes.

Scenario	Pathways (R-HSA)	Spec.	Rxns	Par.	MB
small	1660508, 1660537	3	2	5	0.02
medium	1059683, 109703	40	22	52	0.19
large	111997, 9855142	156	82	214	0.74

4.2. Per-Stage Performance and Scaling

Figure 1 summarizes the pipeline’s resource usage across all three scenarios. Panels (a) and (b) report the scaling of total wall-clock time and peak resident memory with combined input size, while panels (c) and (d) decompose these totals across the individual pipeline stages. Optimization dominates the wall-clock time, whereas the optimization and simulation stages are the most memory-intensive; both time and memory grow steeply from the small to the large scenario.

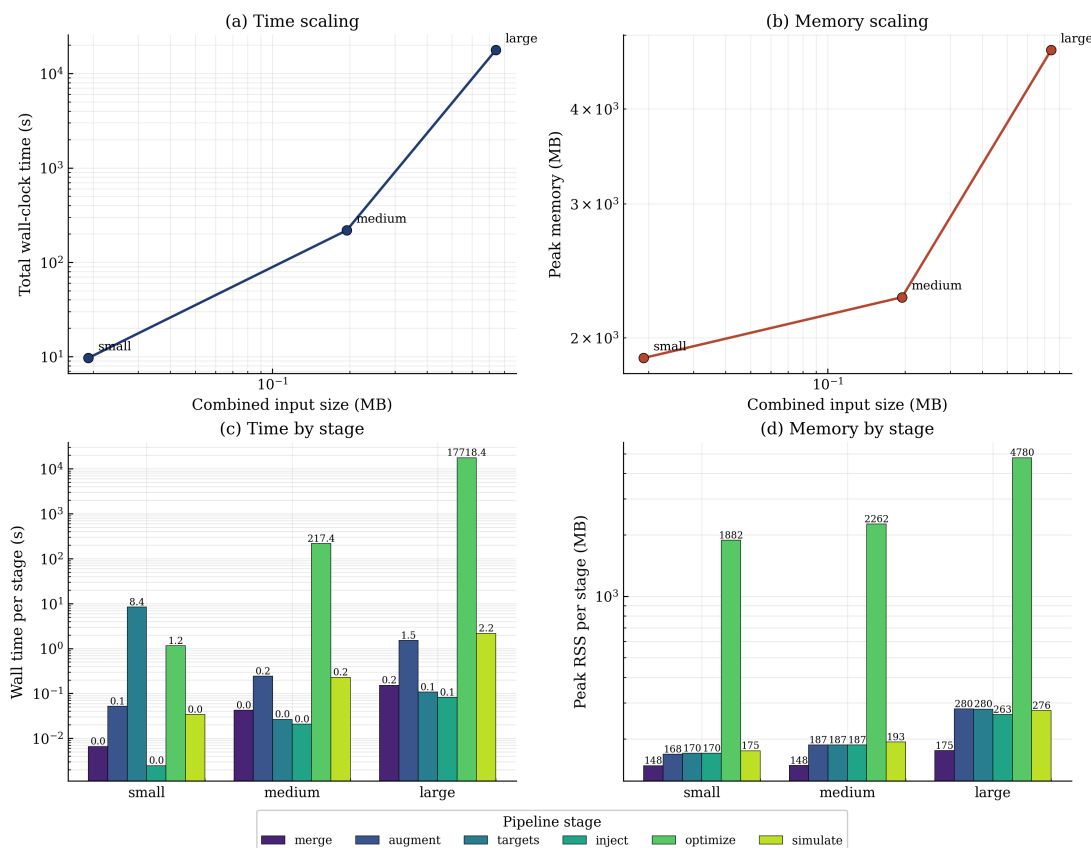


Figure 1. Pipeline performance across the three benchmark scenarios. (a) Total wall-clock time and (b) peak memory as a function of combined input size; (c) wall-clock time and (d) peak resident memory broken down by pipeline stage. All axes use a logarithmic scale.

4.3. Optimization Convergence

Figures 2a to 2c show the optimization loss (Section 3.5) over iterations for each scenario, while Figures 3a to 3c detail the per-species terminal error after fitting.

For the small scenario, the optimizer rapidly converged, triggering the early stopping mechanism at iteration 817 and successfully minimizing the loss to 8.38×10^{-5} , with all species reaching the target steady state within a maximum relative error of 1.5%. Conversely, both the medium and large scenarios exhausted the full 5000 iteration budget without triggering

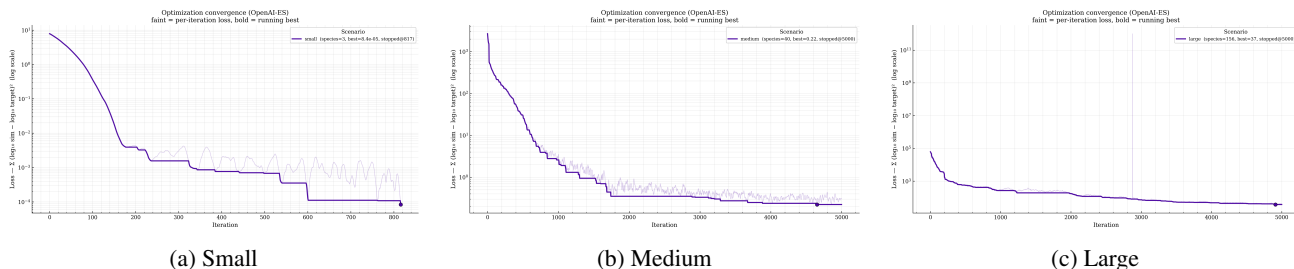


Figure 2. Optimization convergence (OpenAI-ES) for the three benchmark scenarios.

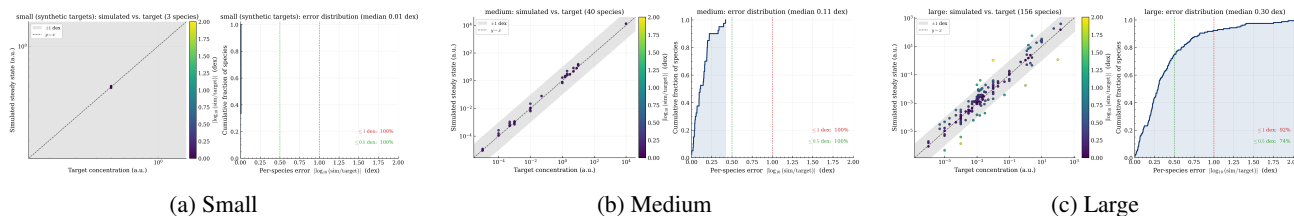


Figure 3. Final error distributions after optimization for the three benchmark scenarios. For each scenario, the left panel shows the log-log scatter plot of simulated versus target steady-state concentrations, while the right panel reports the ECDF of the per-species log-error.

early stopping. The medium scenario achieved a final loss of 0.3124, with 24 out of 40 species (60%) satisfying the 30% relative error tolerance limit. In the large scenario, the final loss plateaued at 41.08, with 35 out of 156 species (22.4%) falling within the tolerance limit, reflecting the heightened difficulty of simultaneously resolving 214 tunable parameters in a highly interconnected biological network.

We additionally observed two methodological pitfalls whose resolution proved essential for reliable convergence and which we report as practical lessons: (i) applying steep Hill regulation to the boundary source and sink reactions induces a near-staircase loss landscape that stalls the optimizer, which is avoided by keeping those terms linear (Section 3.3); and (ii) a clipped relative-error objective saturates for targets that differ by many orders of magnitude, which the logarithmic objective (Section 3.5) resolves.

4.4. Target Generation Quality

While an exhaustive validation of the LLM-derived targets against experimental biological data falls outside the primary scope of this computational methodology, preliminary qualitative checks suggest that the generated target concentrations and compartment volumes are biologically plausible in their respective orders of magnitude. Furthermore, the pipeline’s modular architecture ensures that the baseline LLM utilized in this study can be seamlessly swapped and upgraded with any other domain-expert language model tailored to systems biology.

4.5. End-to-End Validation

To confirm that the optimizer has recovered a true, stable steady state rather than a transient intersection with the target values, we subjected the best-performing parameter sets for all three benchmark models to the extended simulation horizon (Section 3.7). Figure 4 reports, for each scenario, every species’ trajectory normalized by its own target ($s_i(t)/\mu_i$), summarized as a median curve with 25–75% and 10–90% ribbons so that the same representation scales from the 3-species small model to the 156-species large one. In the small scenario, all species reach and hold their targets almost immediately. In the medium scenario the trajectories converge smoothly onto the target line. For the large scenario, the band converges toward the target with a wider residual spread, consistent with the harder 214-parameter fit; crucially, it settles without degrading into long-term oscillations or divergence. This confirms that the derivative-free optimization over the augmented kinetic laws successfully bridges the gap between a static Reactome topology and a dynamically stable model.

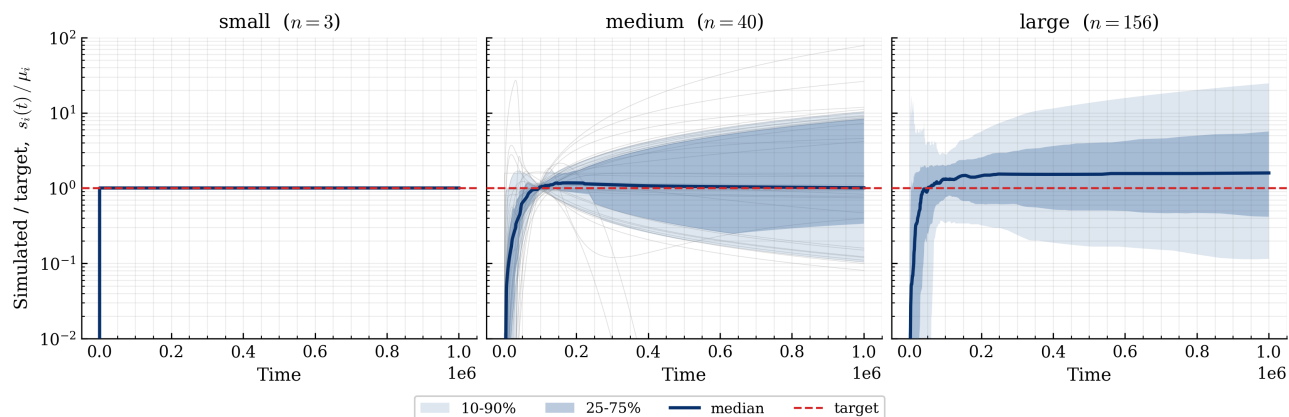


Figure 4. End-to-end validation across the three benchmark scenarios. Each species’ trajectory is normalized by its LLM-derived target, $s_i(t)/\mu_i$; the bold curve is the per-species median, the shaded ribbons span the 25–75% and 10–90% percentiles, and the dashed line marks the target ($= 1$). Faint individual traces are overlaid where legible ($n \leq 40$). All panels share a common log-scaled axis, so the convergence of the small, medium, and large models is directly comparable.

5. Conclusions

We have successfully developed and demonstrated a hybrid workflow that automates the parameterization and enrichment of SBML dynamic models. By utilizing Large Language Models to deduce target biological states from semantic metadata, we bypass one of the most significant data scarcity bottlenecks in computational biology.

References

- Hucka, M., Finney, A., Sauro, H. M., Bolouri, H., Doyle, J. C., Kitano, H., Arkin, A. P., Bornstein, B. J., Bray, D., Cornish-Bowden, A., Cuellar, A. A., Dronov, S., Gilles, E. D., Ginkel, M., Gor, V., Goryanin, I. I., Hedley, W. J., Hodgman, T. C., Hofmeyr, J.-H., Hunter, P. J., Juty, N. S., Kasberger, J. L., Kremling, A., Kummer, U., Le Novère, N., Loew, L. M., Lucio, D., Mendes, P., Minch, E., Mjolsness, E. D., Nakayama, Y., Nelson, M. R., Nielsen, P. F., Sakurada, T., Schaff, J. C., Shapiro, B. E., Shimizu, T. S., Spence, H. D., Stelling, J., Takahashi, K., Tomita, M., Wagner, J., Wang, J., et al. The systems biology markup language (sbml): a medium for representation and exchange of biochemical network models. *Bioinformatics*, 19(4):524–531, 03 2003. ISSN 1367-4803. doi: 10.1093/bioinformatics/btg015. URL <https://doi.org/10.1093/bioinformatics/btg015>.
- Ragueneau, E., Gong, C., Sinquin, P., Sevilla, C., Beavers, D., Gretnier, A., Griss, J., Hogue, G. F. J., Li, N. T., Matthews, L., May, B., Milacic, M., Mohammadi, H., Petryszak, R., Rothfels, K., Shamovsky, V., Stephan, R., Tiwari, K., Weiser, J., Wright, A., Gillespie, M., Wu, G., Stein, L., Hermjakob, H., and D’Eustachio, P. The reactome knowledgebase 2026. *Nucleic Acids Research*, 54(D1):D673–D681, 2026. doi: 10.1093/nar/gkaf1223. PMID: 41251150; PMCID: PMC12807730.
- Salimans, T., Ho, J., Chen, X., Sidor, S., and Sutskever, I. Evolution strategies as a scalable alternative to reinforcement learning, 2017. URL <https://arxiv.org/abs/1703.03864>.
- Welsh, C., Xu, J., Smith, L., König, M., Choi, K., and Sauro, H. M. libroadrunner 2.0: A high-performance sbml simulation and analysis library, 2022. URL <https://arxiv.org/abs/2203.01175>.